

Group 4

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Correspondence Theorem (Artin)

Problem 1. Describe how to tell from the cycle decomposition whether a permutation is odd or even.

Solution. To determine if a permutation is even or odd, if the number of disjoint cycles in a permutation's cycle decomposition is even then the permutation is even. Conversely if the number of disjoint cycles is odd then the permutation is odd. ■

Problem 2. Let H and K be subgroups of a group G .

1. Prove that the intersection $xH \cap yK$ of two cosets of H and K is either empty or else is a coset of the subgroup $H \cap K$.
2. Prove that if H and K have finite index in G then $H \cap K$ also has finite index in G .

Claim 2a. The intersection $xH \cap yK$ of two cosets of H and K is either empty or else is a coset of the subgroup $H \cap K$.

Proof. Suppose $xH \cap yK \neq \emptyset$. Then there exists some $g \in xH \cap yK$ thus $g \in xH$ and $g \in yK$. By definition of left coset, $g = xh_1 = yk_1$ for some $h_1 \in H$ and $k_1 \in K$. Thus $x = gh_1^{-1}$. $\forall s \in xH$, by definition of left coset, for some $h_2 \in H$, $s = xh_2 = gh_1^{-1}h_2$ since $h_1 \in H$, by law of inverses $h_1^{-1} \in H$ and by closure $h_1^{-1}h_2 \in H$. Thus $s = gh_1^{-1}h_2 \in gH$. Thus $xH \subseteq gH$. $\forall t \in gH$, for some $h_3 \in H$ $t = gh_3 = xh_1h_3$ thus $t = xh_1h_3 \in xH$. Thus $gH \subseteq xH$. Since $xH \subseteq gH \subseteq xH$ thus $gH = xH$. The argument is symmetric for $yK = gK$. Thus $xH \cap yK = gH \cap gK$. Every element in $a \in gH \cap gK$ is $a = gm$ for $m \in H$ and $a = gn$ for $n \in K$. Then $gm = gn$, and $m = n$. Thus $m \in H$ and $m \in K$, thus $m \in H \cap K$. Thus $gH \cap gK \subset g(H \cap K)$. $\forall b \in g(H \cap K)$ $b = gm$ for $m \in H \cap K$ thus $m \in H$ and $m \in K$. Thus $b = gm \in gH$ and $b = gm \in gK$ thus $b \in gH \cap gK$. Thus $g(H \cap K) \subset gH \cap gK \subset g(H \cap K)$, thus $g(H \cap K) = gH \cap gK = xH \cap yK$. □

Claim 2b. Prove that if H and K have finite index in G then $H \cap K$ also has finite index in G .

Proof. Since $[G : K]$ is finite then $[H : K \cap H]$ is finite as well, because $H/K \cap H \subset G/K$. Thus by the multiplicative product of indexes $[G : K \cap H] = [G : H][H : K \cap H]$, since $[G : H]$ and $[H : K \cap H]$ are finite then $[G : K \cap H]$ must be finite as well. □

Problem 4. With the notation of the Correspondence Theorem, let H and H' be corresponding subgroups. Prove that $[G : H] = [G' : H']$

Proof. Let H and H' be corresponding subgroups. Since H and H' are corresponding subgroups then $f : G \rightarrow G'$ must also be surjective. Thus since $\ker f \in G$ and f is surjective, then G and $f_*(G) = G'$ are corresponding "subgroups" (groups). Let $K := \ker f$. Thus $|G| = |K||G'|$ and $|H| = |K||H'|$. Thus $|H'| = \frac{|H|}{|K|}$. Divide both sides of $|G| = |K||G'|$ by $|H|$, thus $\frac{|G|}{|H|} = \frac{|K||G'|}{|H|}$. Since $|H'| = \frac{|H|}{|K|}$, then $\frac{|G|}{|H|} = \frac{|G'|}{|H'|}$. By the counting formula and lagrange's theorem, $|G| = [G : H]|H|$ and $|G'| = [G' : H']|H'|$ thus $[G : H] = \frac{|G|}{|H|}$ and $[G' : H'] = \frac{|G'|}{|H'|}$. Thus because $\frac{|G|}{|H|} = \frac{|G'|}{|H'|}$, then $[G : H] = [G' : H']$. $\square$

Problem 5. With reference to the homomorphism $S^4 \rightarrow S^3$ described in Example 2.5.13, determine the six subgroups of S^4 that contain K .

Proof. The homomorphism $f : S^4 \rightarrow S^3$ in Example 2.5.13 is a surjective homomorphism. By the Correspondence Theorem, each subgroup in S^4 which contain K is in a bijective correspondence with a subgroup of S^3 . Because $|S_3| = 6 = |D_3|$ and S_3 and D_3 share the same generators and rules there is a surjective and injective homomorphism between the groups so the groups are isomorphic. So each subgroup of D^3 has a isomorphic equivalent in S_3 . The subgroups of D_3 are $1, C_3, \{1, y\}, \{1, xy\}, \{1, x^2y\}, D_3$. Thus the subgroups of S_3 are $1, \langle x \rangle, \{1, y\}, \{1, xy\}, \{1, x^2y\}, S_3$ where $x = (123)$ and $y = (12)$. Thus by the correspondence Theorem the subgroups of S^4 containing K , are $f^*(1), f^*(\langle x \rangle), f^*(\{1, y\}), f^*(\{1, xy\}), f^*(\{1, x^2y\}), f^*(S_3)$. Thus they are $\{1, (12)(34), (13)(24), (14)(23)\}, \langle (312) \rangle, \{1, (1, 2)\}, \{1, (312)(12)\}, \{1, (312)^2(12)\}, D_4$ $\square$

Product Groups (Artin)

Problem 1. Let x be an element of order r of a group G , and let y be an element of G' of order s . What is the order of (x, y) in the product group $G \times G'$?

Proof. Order of (x, y) is $n = \text{lcm}(r, s)$. $\square$

Problem 3. Prove that the product of two infinite cyclic groups is not infinite cyclic.

Proof. Any infinite cyclic group is isomorphic to $\mathbb{Z}$. and $\mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^2$, by definition, which is generated by $(1, 0), (0, 1)$ and thus is not infinite cyclic group. Thus product of two infinite cyclic groups is not infinite cyclic. $\square$

Problem 4. In each of the following cases, determine whether or not G is isomorphic to the product group $H \times K$.

1. $G = \mathbb{R}^\times, H = \{\pm 1\}, K = \{\text{positive real numbers}\}$
2. $G = \{\text{invertible upper triangular } 2 \times 2 \text{ matrices}\}, H = \{\text{invertible diagonal matrices}\}, K = \{\text{upper triangular matrices with diagonal entries } 1\}$
3. $G = Cx, H = \text{unit circle}, K = \{\text{positive real numbers}\}.$

- Solution.* 1. $G \cong H \times K$. $f : H \times K \rightarrow G$ takes $(\pm 1, r) \rightarrow \pm r$ where $r \in \{\text{positive real numbers}\}$.
 $g : G \rightarrow H \times K$ which takes $x \rightarrow (\frac{x}{|x|}, |x|)$. f is a homomorphism and $f \circ g = id_{H \times K}$ and $g \circ f = id_{\mathbb{R}^\times}$
2. $G \cong H \times K$. Every invertible upper triangular matrix $M = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ where $ac \neq 0$ can be written as the product $M = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} = DU$. $D \in \{\text{invertible diagonal matrices}\}$ and $U \in \{\text{upper triangular matrices with diagonal entries 1}\}$. Since they can all be written as this product the map from $(H \times K) \rightarrow G$ is the natural map which is both bijective and homomorphism. Thus $D \cong H \times K$
3. H represents a rotation and K represents scaling. $f : H \times K \rightarrow C^\times$ s.t. $(e^{i\theta}, k) \rightarrow ke^{i\theta}$ and since every complex number is given by a unique pair of angle and magnitude the function is bijective. Furthermore f is a homomorphism. Thus $C^\times \cong H \times K$ ■

Problem 9. Let H and K be subgroups of a group G . Prove that the product set HK is a subgroup of G if and only if $HK = KH$.

Proof. ($\implies$) : Let H and K be subgroups of a group G . HK is a subgroup of G . By definition of HK , for all $x \in HK$, $x = hk$ for $h \in H$ and $k \in K$. Since HK is also a subgroup of G , $x^{-1} \in HK$, $x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH$. Thus $\forall x \in HK$, $x^{-1} \in KH$, thus since $x^{-1} \in HK$ then $x \in KH$. Thus $HK \subseteq KH$. Similarly for $kh \in KH$, $he = h \in HK$ and $ek = k \in HK$ since HK is closed $hk \in HK$ thus $KH \subseteq HK$. Thus $HK = KH$.

($\impliedby$) : Let $HK = KH$. Thus by definition $\forall k \in K$ and $\forall h \in H$, $hk = kh$.

1. **Closed:** Let $x, y \in HK$ then by definition $x = h_1k_1$ and $y = h_2k_2$, $xy = h_1k_1h_2k_2$ however since $k_1h_2 = h_2k_1$ then $xy = h_1h_2k_1k_2$ thus $xy \in HK$.
2. **Inverses:** Let $x \in HK$, then $x = hk = kh$. $k^{-1}h^{-1} \in KH = HK$. Thus $xk^{-1}h^{-1} = hkk^{-1}h^{-1} = e$ and $k^{-1}h^{-1}x = k^{-1}h^{-1}hk = e$. Thus x has a inverse $h^{-1}k^{-1} = k^{-1}h^{-1} \in KH = HK$
3. **Identity:** Since K and H are subgroups, $e \in K$ and $e \in H$ thus $ee = e \in KH$. Thus $HK = KH$ have a identity.

Thus $HK = KH$ are a subgroup in G . □

Keel Problems

Problem 1a. Give natural injective homos from S and T to $S \times T$. So we can think of S and T as subgroups of $S \times T$.

Solution. The natural injective homomorphism $f : S \rightarrow S \times T$ is $\forall s \in S$ $f(s) = (s, e)$ and the natural injective homomorphism $g : T \rightarrow S \times T$ is $\forall t \in T$ $g(t) = (e, t)$. Suppose for $x \in S$, $f(x) = (x, e) = e = (e, e)$, then $x = e$ thus $\ker f = \{e\}$ and f is injective. Similarly, for $x \in T$, $g(x) = (e, x) = e = (e, e)$, then $x = e$ thus $\ker g = \{e\}$ and g is injective. ■

Claim 1b. Show that they commute with each other (ie $st = ts$ for s in S and t in T , where this product is taking place in $S \times T$).

Proof. $st = (s, e_t) \cdot (e_s, t) = (s * e_s, e_t * t) = (s, t)$. By property of identities in groups, $(s, t) = (e_s * s, t * e_t) = (e_s, t) \cdot (s, e_t) = ts$ $\square$

Claim 1c. $S \cap T = \{e\}$

Proof. Let $x \in S \cap T$, then $x \in S$ and $x \in T$. Thus $x = (x, e_t) = (e_s, x)$. Thus $(x, e_t) = (e_s, x)$ thus $x = e_s$ and $x = e_t$. Thus $x = (e_s, e_t) = e$. Thus the only element in $S \cap T$ is e . $\square$

Claim 1d. $ST = TS = S \times T$

Proof. $\forall x \in ST$ by definition $x = st$. By 1b, $x = st = ts \in TS$. Thus $\forall x \in ST, x \in TS$. $ST \subseteq TS$. $\forall y \in TS$ by definition $y = ts = st \in ST$. Thus $\forall y \in TS, y \in ST$. Thus $TS \subseteq ST$ and $ST = TS$. $\forall x \in ST, x = st = (s, e_t) \cdot (e_s, t) \in S \times T$. Thus $\forall x \in ST, x \in S \times T$. Thus $ST \subseteq S \times T$. For all $a \in S \times T, a = (s, t) = (s * e_s, e_t * t) = (s, e_t) \cdot (e_s, t) = st \in ST$. Thus $S \times T \subseteq ST$. Thus $ST = S \times T = TS$. $\square$

Problem 2. Now we try to figure out how to do "the opposite", we start with a group G and subgroups S, T . Let $m : S \times T \rightarrow G$ be the natural map, ie $(s, t) \rightarrow st$. We want to know when this map is an isomorphism of groups.

Claim 2a. Show that m is a homomorphism iff S and T commute, ie $st = ts$ for all s in $s \in S$ and $t \in T$.

Proof. Let G be a group with subgroups S, T . Let $m : S \times T \rightarrow G$ be the natural map, ie $(s, t) \rightarrow st$.

($\implies$): Let m be a homomorphism. Then $\forall x \in S \times T$, by definition $x = (s, t)$ for $s \in S$ and $t \in T$. $x = (s, t) \cdot (s, t) = (ss, tt)$. $m(xx) = m[(ss, tt)] = sstt$. Since m is a homomorphism, $m(xx) = m(x)m(x) = stst$. Thus $sstt = stst$, and $s^{-1}ststt^{-1} = s^{-1}ssttt^{-1}$ is $ts = st$. Since x can be any element of $S \times T$, this is true for all $s \in S$ and $t \in T$

($\impliedby$): Suppose $st = ts$ for all s in $s \in S$ and $t \in T$. Then $\forall x, y \in S \times T$, by definition $x = (s_1, t_1)$ and $y = (s_2, t_2)$ s.t $s_1, s_2 \in S$ and t_1, t_2 . By definition of $m(x) = s_1t_1$ and $m(y) = s_2t_2$. $xy = (s_1s_2, t_1t_2)$ by definition of multiplication in the product group. $m(xy) = m[(s_1s_2, t_1t_2)] = s_1s_2t_1t_2$. Since $st = ts$ for all s in $s \in S$ and $t \in T$, $s_2t_1 = t_1s_2$ thus $m(xy) = s_1s_2t_1t_2 = s_1t_1s_2t_2 = m(x)m(y)$. Thus $\forall x, y \in S \times T, m(xy) = m(x)m(y)$. Thus m is a homomorphism. $\square$

Claim 2b. $m_*(S \times T) = ST$

Proof. $\forall x \in S \times T$, by definition $x = (s, t)$ for $s \in S$ and $t \in T$. $m(x) = m((s, t)) = st \in ST$. Thus $m_*(S \times T) \subseteq ST$.

$\forall y \in ST$, by definition $y = st$ for $s \in S$ and $t \in T$. $\exists x \in S \times T$ s.t $x = (s, t)$. Thus $m(x) = st = y$. Thus $y \in m_*(S \times T)$. Thus $ST \subseteq m_*(S \times T)$. Thus $ST = m_*(S \times T)$ because $m_*(S \times T) \subseteq ST \subseteq m_*(S \times T)$ $\square$

Claim 2c. Now suppose S and T do commute, so m is a homomorphism. Show that its kernel is naturally identified with $S \cap T$.

Proof. Suppose $x \in \ker m$. Thus $m(x) = e$. $x = (s, t)$. Thus $m((s, t)) = st = e$. This is only possible if $s = t^{-1}$ or if $s = e$ and $t = e$. If $s = t^{-1}$, then $s = t^{-1} \in T$ thus $S \subseteq T$, thus $x \in S \cap T$. If $s = e$ and $t = e$, $e \in S$ and $e \in T$, thus $x = e$ and $x \in S \cap T$. Thus $\ker m \subseteq S \cap T$. Suppose $x \in S \cap T$ s.t $x \in S \times T$ s.t $a \in S \cap T$ and $b \in S \cap T$. Then $m(x) = ab$ $\square$

Claim 2d. m is an isomorphism iff all the following hold: S and T commute, $S \cap T = \{e\}$, $ST = G$.

Proof. ($\Leftarrow$): Suppose $S \cap T = \{e\}$ and $ST = G$. Then it's a bijection by 2e. According to 2a, it is a homomorphism iff $ST = TS$ or S and T commute. Thus m is a bijection and a homomorphism and thus by definition of isomorphism it is an isomorphism.

($\Rightarrow$): Suppose m is an isomorphism. By definition it's a bijection and an isomorphism. By 2e, $S \cap T = \{e\}$ and $ST = G$. By 2a, since m is a homomorphism $ST = TS$ and thus S and T commute. $\square$

Claim 2e. Show that m is a bijection iff $S \cap T = e$ and $ST = G$.

Proof. ($\Rightarrow$): Suppose m is a bijection. Then it is injective and surjective.

Thus $\forall a, b \in S \times T$, by definition $a = (s_1, t_1)$ and $b = (s_2, t_2)$ if $m(a) = m(b)$, then $a = b$ by definition of injective. Assume the contrary, $\exists x \in S \cap T$ s.t. $x \neq e$. Then $x \in S$ and $x^{-1} \in S$ because S is a subgroup. Thus $m((x^{-1}, x)) = x^{-1}x = e$. Similarly $x \in T$ and $x^{-1} \in T$, thus $m((x, x^{-1})) = xx^{-1} = e$. Thus $m((x, x^{-1})) = m((x^{-1}, x))$, but $(x, x^{-1}) \neq (x^{-1}, x)$. This is a contradiction with the fact, m is injective. Thus $S \cap T = \{e\}$.

Since m is surjective, $m_*(S \times T) = G$ by definition of image. By 2b, we know $m_*(S \times T) = ST$. Thus $ST = G$.

($\Leftarrow$): Suppose $ST = G$ and $S \cap T = e$. By 2b we know $ST = m_*(S \times T)$. Thus $m_*(S \times T) = G$, by definition of image and surjective functions, m must be surjective. Let $x = (s_1, t_1), y = (s_2, t_2) \in S \times T$ s.t. $m(x) = s_1t_1 = s_2t_2 = m(y)$. Then $s_1^{-1}s_1t_1t_2^{-1} = s_1^{-1}s_2t_2t_2^{-1}$ which is $t_1t_2^{-1} = s_1^{-1}s_2$. $t_1t_2^{-1} \in T$ and $s_1^{-1}s_2 \in S$. But since $t_1t_2^{-1} = s_1^{-1}s_2$, then $t_1t_2^{-1} = s_1^{-1}s_2 \in S \cap T$. However since e is the only element of $S \cap T$, $t_1t_2^{-1} = s_1^{-1}s_2 = e$. Thus $t_1 = t_2$ and $s_1 = s_2$, thus $x = y$. Thus m is injective. $\square$

Problem 6. Remark: if we are as in 5) and $m : S \times T \rightarrow G$ is a bijection, but S and T don't commute, this means we get a new group law on the set $S \times T$ – namely the group law from G . This will be very important when we classify groups of small order later in the class. There are lots of examples of subgroups S and T of G where $S \cap T = e$ and $ST = G$, but S and T do not commute. Show this is true for $G = D_n, S = C_n$ and $T = \langle y \rangle$ (the subgroups generated by a reflection).

Proof. As seen in the previous homework, $\forall g \in D_n, g = x^a y^b$ for $a \in [0, \dots, n-1]$ and $b \in [0, 1]$. Thus $g = x^a y^b \in ST$. Thus $G \subseteq ST$. However $ST \subset G$ by definition so $G = ST$. By the rules of D_n reflections and rotations don't commute. Suppose $g \in S \cap T$, then $g \in S = \{e, x, \dots, x^n\}$ and $g \in T = \{e, y\}$. Thus $g = x^a = y^b$ for some $a, b \in \mathbb{Z}$. Thus $x^a y^{-b} = x^a y^b = e$ this is only true if $a = b = 0$. Thus $g = e$ and $S \cap T = e$. Thus there is a bijection from $S \times T \rightarrow G$. $\square$